

Firefighters' Safety During Industrial Fire Scenarios in Oil & Gas Facilities

Focus on Thermal Hazards and Protective Ensembles (NFPA 1971)



Author:

Antar Rezaiguia “HSE Master Degree (Firefighting Specialist, Oil & Gas Sector)”

Reviewed by:

Kebbech Moussa “Head of intervention Section at SONATRACH with 26 years of experience in oil & gas operations and emergency response”

Abib Ben Chergui “HSE Intervention Forman at SONTRACH with 20 years of experience”

Email: antar.rezaiguia@yahoo.com

Date: December 2025

Contents

1. Introduction, Scope and Limits:.....	3
Scope:.....	4
Limits:.....	4
2. Fire.....	4
Fire	4
Type of fires in oil & gas industry	4
3. Fire consequences.....	6
Flames, Heat flux.....	6
Thermal radiations levels	8
Smoke and Toxic Gas Generation.....	9
Other consequences.....	9
4. Firefighters Vulnerability.....	10
Engulfment by fire (inside the flame)	10
Thermal Radiation from the Fire (Outside the Flame).....	10
Time and Distance	11
Thermal Dose (TDU) & Probit Models	13
Lethality Thresholds	15
5. Protective ensemble for firefighting agents	16
NFPA 1971 Requirements	17
Main Performance Parameters to Verify According to NFPA 1971 (2018)	20
Other Standards	21
6. Conclusion.....	22

Introduction, Scope and Limits:

With 495 oil and gas fatalities, 401 from fire and explosion accidents, and 22 emergency and rescue firefighter deaths out of 14,945 total US cases from 2009-2022 per my OSHA data analysis, fires in the oil and gas industry emerge as the biggest risk. The International Association of Oil & Gas Producers (IOGP) reported 32 worker fatalities in 2024, with explosions, fires, or burns as a major cause. The NFPA recorded 62 US firefighter deaths in 2024, 65% due to suppression-related stress, while the USFA noted 72 fatalities, including 42 from overexertion. Driven by flammable hydrocarbons, high-pressure systems, and volatile environments, these incidents—escalating into jet fires, pool fires, or explosions—threaten responders during suppression. This data highlights the urgent need to focus on fire safety in this sector.

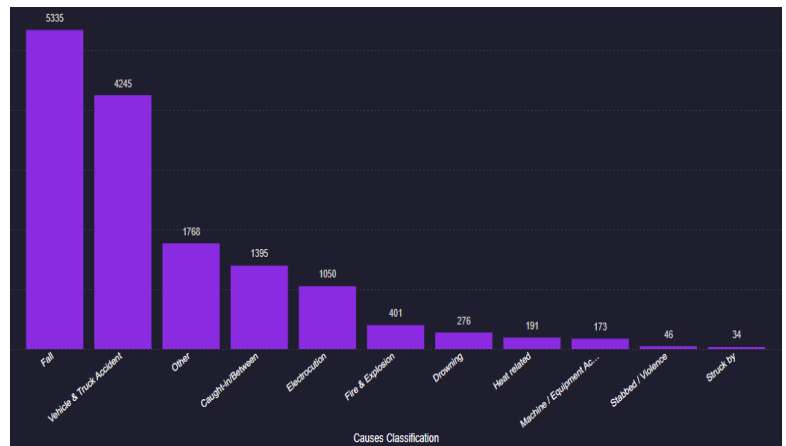


Figure 1 USA OSHA fatalities analysis -chart of causes classification-

With 17 firefighter deaths, 132 injuries, and over 5,000 evacuations, the Matanzas oil storage facility explosion on August 5, 2022, in Cuba, highlights the dire risks of fire suppression in oil and gas. A lightning strike ignited an oil tank, sparking a fire that spread to four tanks and triggered multiple explosions. This incident underscores the direct and severe dangers faced by firefighting teams in such volatile environments.



Figure 2 Matanzas oil storage facility explosion

Scope:

This research focuses exclusively on fire incidents within the oil and gas industry, emphasizing safety during suppression scenarios. It draws on standards like NFPA 1971 for protective ensembles and IOGP safety data for incident insights, aiming to analyze fire types, consequences, human vulnerabilities, and protective measures for fire agents. It's rare to find documents that delve into these specific details of firefighter safety during suppression in oil and gas, making this study a unique contribution.

Limits:

This document covers only firefighter safety related to fires in the oil and gas industry, focusing on heat and flames hazards and excluding the safety of other workers, other types of accidents (e.g., chemical spills or equipment failures unless they ignite fires), and incidents in other industries. It excludes fire prevention strategies, post-incident recovery, and broader emergency responses.

Fire

Fire

Fire is a rapid exothermic oxidation reaction producing heat, light, and flames, driven by the fire triangle (fuel, oxygen, ignition source).

In the oil and gas industry, this process is amplified by the abundance of flammable hydrocarbons (e.g., crude oil, natural gas) and high-pressure conditions, making fires a dominant hazard during operations and suppression efforts.



Figure 3 fire in oil & gas field

Type of fires in oil & gas industry

Pool Fires:

- *Description:* Fires involving a pool of flammable liquid (e.g., oil spill) burning on a surface. Common in storage tank leaks or ruptured pipelines.

- *Characteristics:* Steady burning with high heat flux (up to 150 kW/m² near the source), large flame heights, and potential for rapid spread.
- *Example:* Tank farm fires, like Matanzas (2022), where fuel pooling fueled the blaze.

Jet Fires:

- *Description:* High-velocity fires from pressurized leaks (e.g., gas pipelines or well blowouts).
- *Characteristics:* Intense, directional flames with heat flux exceeding 200 kW/m² close to the source, rapid flame spread, and risk of equipment failure.
- *Example:* Gas pipeline ruptures in midstream operations.

Flash Fires:

- *Description:* Brief, explosive ignition of a vapor cloud (e.g., gas release igniting).
- *Characteristics:* Short duration (seconds), widespread flame front, and high risk of burns or asphyxiation from toxic smoke.
- *Example:* Offshore platform gas leaks igniting during maintenance.

Ball Fire:

- *Description:* A rare, spherical fireball resulting from the sudden release and ignition of a large volume of flammable vapor or gas (e.g., during a catastrophic tank rupture).
- *Characteristics:* Extremely high heat intensity (up to 300 kW/m²), short-lived (5-10 seconds), and capable of causing severe burns over a wide area.
- *Example:* Potential outcome of a major oil tank explosion under pressure.



Figure 4- AI image for types of fires in oil and gas

Fire consequences

Flames, Heat flux

Flames in oil and gas fires, driven by hydrocarbons, exhibit significant vertical and horizontal reach, posing direct threats to firefighters. The CCPS Guidelines model flame lengths for pool fires using the equation

$$L/d = 42 Q^{2/5}$$

where:

L is length,

d is diameter,

Q is heat release rate

estimating a 10-meter diameter pool fire to produce flames up to 20-25 meters high under calm conditions.

HSE UK's Fire and Explosion Guidance specifies that jet fires from pressurized releases (e.g., 5-10 bar) can extend 30-50 meters horizontally, depending on release velocity and wind.

IOGP Report 434-14, based on upstream incident data, notes flash fires can spread flames 20-40 meters in seconds from a vapor cloud ignition, while ball fires from tank ruptures may project flames up to 50-70 meters radially.

API RP 521 advises that flame impingement distances for equipment exposure align with these ranges, recommending a 30-meter buffer for jet fires to avoid structural failure.

The SFPE Handbook refines this with empirical correlations, suggesting pool fire flame heights increase by 50% in windy conditions (e.g., 30-40 meters for a 20-meter pool).

Distances: The reach of flames dictates safe operational zones for firefighters. Pool fires typically extend 20-40 meters vertically, requiring a 25-50 meters standoff to avoid direct contact, with HSE guidance emphasizing wind-driven tilt adding 10-15 meters laterally. Jet fires demand a 50-70 meters exclusion zone due to their directional reach, as per API RP 521, with IOGP data confirming fatalities within 30 meters during upstream incidents. Flash fires, with a 20-40 meters flame front, necessitate immediate retreat beyond 50 meters, while ball fires can engulf areas up to 70 meters, aligning with SFPE's rapid expansion models. These distances, critical during suppression, inform dynamic safety perimeters and NFPA 1971 gear deployment, with OSHA PSM requiring their integration into emergency planning.

Table 01: flame distances

Fire Type	Flame Distance (m)	Key Factors Influencing Reach	Source/Reference
Pool Fire	20-40 (vertical); +10-15 (lateral with wind)	Pool diameter, wind speed	CCPS Guidelines; HSE Guidance; SFPE Handbook
Jet Fire	30-50 (horizontal); up to 70 (with wind)	Pressure, release velocity	API RP 521; IOGP 434-14
Flash Fire	20-40 (radial spread)	Vapor cloud size, ignition	HSE Fire Guidance; SFPE
Ball Fire	50-70 (radial)	Tank pressure, fuel volume	IOGP 434-14; SFPE Handbook

Heat flux

Fires in oil and gas fields, such as those from hydrocarbons, generate intense convective and radiative heat transfer, quantified as heat flux (kW/m²).

The CCPS Guidelines detail how pool fires from spilled crude can produce fluxes up to 150 kW/m² at the flame base, while jet fires from pressurized gas releases exceed 200 kW/m², leading to rapid material degradation and escalation risks.

HSE UK’s Fire and Explosion Guidance emphasizes that in chemical processes, heat flux drives domino effects like vessel rupture, with firefighters exposed to fluxes causing immediate burns if unprotected.

IOGP Report 434-14 notes that sustained exposure above 10 kW/m² in upstream incidents contributes to 25% of burn-related fatalities, aligning with API RP 521's fire sizing criteria for relief systems to mitigate heat input.

The SFPE Handbook integrates these by modeling flame emissivity and view factors for accurate flux prediction, recommending distance-based barriers for suppression teams. In practice, these fluxes necessitate NFPA 1971-compliant gear, but exceedances during prolonged operations heighten dehydration and fatigue risks, as per OSHA PSM integration.

Table 02: Heat flux values

Heat Flux Level (kW/m ²)	Effect on Exposed Skin (Unprotected)	Typical Fire Type in Oil & Gas	Source/Reference
1-4	No harm for long exposure; mild discomfort	Ambient radiation from small leaks	CCPS Guidelines; SFPE Handbook
4.7-12.5	Pain after 10-20s; first-degree burns in 30s	Pool fire periphery	HSE Fire Guidance; IOGP 434-14
12.5-25	Second-degree burns in 20s; 1% fatality in 10s	Jet fire at 10-15m	API RP 521; CCPS Probit Data
>25	Immediate severe burns; 100% fatality in 1 min	Ball or flash fire core	SFPE Handbook; OSHA PSM

Thermal radiations levels

Thermal radiation, the dominant consequence in large-scale fires, propagates heat without direct contact, impacting firefighters at standoff distances.

The CCPS Guidelines classify radiation levels with thresholds: 1.6 kW/m² for no harm, escalating to 35 kW/m² for 100% second-degree burns in 20s.

HSE UK's HSG143 and Fire Guidance recommend assessing radiation using point-source models for chemical plants, where levels above 12.5 kW/m² trigger mandatory evacuation zones.

IOGP 434-14 provides oil-specific criteria, noting 4.7 kW/m² as the pain threshold after 20s, with 25 kW/m² causing spontaneous clothing ignition, based on upstream incident data. API RP 521 integrates this for facility design, advising radiation limits to protect personnel during fire response, while the SFPE Handbook offers equations for radiant fraction (e.g., 0.2-0.4 for hydrocarbons) to calculate exposure radii.

In HSE engineering for oil and gas, these levels inform safe approach distances, reducing risks of thermal injury during suppression, as OSHA PSM requires quantitative evaluation for flammable inventories.

Table 03: Thermal radiations levels

Thermal Radiation Level (kW/m ²)	Human Effect (Exposure Time)	Application in Oil & Gas	Source/Reference
1.6	No harm (indefinite)	Safe public exposure	CCPS Guidelines; HSE Guidance
4.7	Pain after 20s	Escape limit	IOGP 434-14; SFPE Handbook
12.5	100% second-degree burns (20s); 1% fatality (10s)	Equipment damage zone	API RP 521; CCPS
25	Clothing ignition (immediate); 100% fatality (1 min)	Lethal near fire	HSE HSG143; IOGP
35	100% fatality (20s)	Immediate retreat zone	SFPE Handbook; OSHA PSM

Smoke and Toxic Gas Generation

Combustion of hydrocarbons produces dense smoke and toxic gases (e.g., carbon monoxide, hydrogen sulfide, and benzene), with concentrations escalating in confined or poorly ventilated areas. The CCPS Guidelines note that smoke opacity can reduce visibility to near zero within 50 meters of a large pool fire.

HSE UK guidance highlights H₂S levels exceeding 100 ppm as immediately dangerous to life.

Impact on Firefighters: Inhalation risks lead to asphyxiation or acute poisoning, with IOGP data linking 15% of 2024 upstream fatalities to toxic exposure. *The Matanzas incident (2022) reported respiratory injuries from smoke.*

Other consequences

Explosion and Overpressure Effects:

Vapor cloud explosions (VCEs) or boiling liquid expanding vapor explosions (BLEVEs) can follow ignition, generating overpressures up to 0.2-1 bar within 50-100 meters, per API RP 521. The SFPE Handbook models BLEVE blast radii at 100-200 meters for large storage tanks.

Impact on Firefighters: Overpressure causes eardrum rupture or structural collapse, with HSE noting 10% of chemical plant injuries from secondary blasts.

- **Equipment and Structural Damage:**

Intense heat and flames can weaken steel structures (failure at 500-600°C) and rupture pipelines, as outlined in CCPS Guidelines for domino effects. IOGP Report 434-14 cites equipment failure as a precursor to 20% of fire escalations.

Impact on Firefighters: Falling debris or secondary fires endanger responders, with API RP 521 recommending 30-50 meters buffers from compromised assets.

- **Environmental Contamination:**

Firefighting runoff (e.g., foam, oil residues) and unburned hydrocarbons contaminate soil and water, with HSE UK estimating 100-500 liters of pollutant release per large fire. The SFPE Handbook notes persistent air pollution from smoke plumes.

Impact on Firefighters: Indirect exposure to contaminated areas post-suppression increases long-term health risks (e.g., carcinogens).

Firefighters Vulnerability

Engulfment by fire (inside the flame)

A person momentarily and only partially exposed directly to fire is most likely to suffer pain and non-fatal burns. A person fully or substantially engulfed by fire can be considered to become a fatality.

For engulfment due to jet fire, pool fire or fireball: – 100% lethality for workers or members of the public.

For engulfment due to flash fire: – 100% lethality for members of the public. – 50% to 100% lethality, depending on ease of escape, for workers wearing fire resistant clothing.

For firefighters engulfment occurs when they are within the flame envelope (e.g., 20-40m for pool fires, 30-50m for jet fires).

The CCPS Guidelines estimate heat fluxes up to 300 kW/m² inside ball fire cores, causing instantaneous burns.

HSE UK notes jet fire engulfment as a leading cause of upstream fatalities.

Second- and third-degree burns within 5-10 seconds, with IOGP data reporting 25% of 2024 burn injuries from engulfment.

Thermal Radiation from the Fire (Outside the Flame)

The effects of thermal radiation depend primarily on the thermal radiation flux and the duration of exposure. However, the type of clothing worn, the ease of sheltering, and the individual exposed will also have an influence.

Thermal radiation poses a critical, non-contact threat to firefighters during oil and gas fire suppression. Unlike convective engulfment, radiation penetrates protective ensembles over time and distance, inducing cumulative injury even at standoff positions. This section, grounded in NFPA 1971 (2018), CCPS Guidelines, HSE UK, IOGP 434-14, API RP 521, and SFPE Handbook, quantifies exposure levels, human response, and safe operational limits specifically for firefighters operating outside the flame in proximity firefighting scenarios.

Table 04: Radiant Flux effects

Radiant Flux (kW/m ²)	Effect on Unprotected Skin	Effect on NFPA 1971-Geared Firefighter	Typical Distance from Fire	Source
1.6	No harm (indefinite)	No thermal load; baseline ambient	>50 m	CCPS, HSE
2.0	NFPA 1971 proximity threshold	Safe for indefinite wear	40–50 m	NFPA 1971
4.7	Pain after 20 s	Safe with aluminized proximity suit (>30 min)	25–35 m	IOGP 434-14, SFPE
10.0	2nd-degree burn in 30 s	TPP ₅₀ = 30 s (NFPA 1971 minimum)	15–25 m	CCPS, API RP 521
12.5	1% fatality in 10 s; 100% 2nd-degree in 20 s	Gear failure risk after 20–25 s	10–20 m	HSE, CCPS
25.0	Clothing ignition; 100% fatality in 1 min	Immediate retreat required	<10 m	SFPE, IOGP
35.0	100% fatality in 20 s	Lethal even with proximity gear	<8 m	API RP 521

Time and Distance

Time and distance are the two most controllable variables determining firefighter survival during oil and gas fire suppression. NFPA 1971 (2018) defines *proximity firefighting* as operations where radiant heat flux exceeds 2.0 kW/m² without direct flame contact — the dominant exposure mode outside the flame envelope. This section integrates NFPA 1971, CCPS Guidelines, HSE UK, IOGP 434-14, API RP 521, and SFPE Handbook to provide quantitative time-distance thresholds, safe exposure durations, and operational zoning for firefighters in oil and gas environments.

Table 05: Time-Distance Matrix for Oil & Gas Fires (NFPA 1971-Geared Firefighters)

Fire Type	Flame Size	Distance (m)	Radiant Flux (kW/m ²)	Max Safe Time (NFPA 1971 TPP ≥35)	Action Required
Pool Fire	10 m diameter	10	45	<10 s	Immediate retreat
		20	12.5	23 s	Final suppression push
		30	6.0	49 s	Primary attack zone
		50	2.5	>2 min	Safe staging
Jet Fire	40 m horizontal	15	60	<8 s	No approach
		30	20	14 s	Remote cooling only
		50	8.0	37 s	Boundary control
Ball Fire	50 m radius	20	70	<7 s	Lethal zone
		40	18	16 s	Evacuation trigger
		60	9.0	33 s	Outer perimeter

Table 06: Operational Time-Distance Zones (HSE UK & IOGP 434-14)

Zone	Distance from Flame	Flux (kW/m ²)	Max Exposure Time	Firefighter Action	Command Decision
Red (Lethal)	0–20 m	>25	<15 s	No entry	Evacuate immediately
Orange (High Risk)	20–30 m	12.5–25	15–30 s	Brief attack only	Rotate every 20 s
Yellow (Moderate Risk)	30–50 m	4.7–12.5	30 s – 2 min	Primary suppression	Monitor TDU
Green (Safe)	>50 m	<4.7	Indefinite	Staging, logistics	Command post

NFPA 1971, IOGP, and HSE converge on:

- **≥30 m standoff** for primary suppression
- **≤30 s exposure** at 10 kW/m²

Thermal Dose (TDU) & Probit Models

Thermal radiation dose units (TDU) are an appropriate way of combining levels of radiation with exposure times:

$$\text{Dose (TDU)} = Q^{4/3} t \text{ where :}$$

Q: is incident thermal flux (kW/m²), and

t: is the duration of exposure (s).

Thermal dose units (TDU) thus have the units (kW/m²)^{4/3}s

The dose (TDU) required to reach selected levels of harm to humans are reported by the UK HSE [8]; for both infrared radiations associated with hydrocarbon fires and ultraviolet radiation associated with nuclear explosions. Table .., replicated from [8], provides the more appropriate, lower, values associated with infrared radiation together with the source references.

Table 7: Thermal radiation doses for selected harm levels

Thermal Radiation Range Minimum	Dose (TDU)	Reference
Pain	108 – 12786 – 103	DNV/Scandpower [10]HSL [11]
Significant Injury / First Degree Burns	$\leq 8013080 - 130$	Metha et al. [12]Tsao & Perry (1979) [13]HSL [11]
Second Degree Burns / 1% Lethality for Average Clothing	270 – 310≈ 350730290	Stoll & Green [14]Metha et al. [12]Arnold et al. [15]HSL [11]
Third Degree Burns / 50% Lethality for Average Clothing	≈ 5002000 – 3000870 – 2600	Metha et al. [12]DNV/Scandpower [10]HSL [11]

Probits

A probit approach can be used instead of the dose criteria described in Section before. In this approach, the probability of fatality, or probit, takes the form

$$Y = a + b \ln (\text{TDU})$$

- Standard Probit Equation for Thermal Radiation (Unprotected Skin)

$$Y = -36.38 + 2.56 \times \ln (\text{TDU})$$

- The literature details various values for parameters a and b. Table... provides the most commonly used values of a and b and the dose required to result in 1% and 50% lethality or probits (referenced in [8]).

Table 08: TDU 1% and 50% lethality

Source	a	b	1% Lethal Dose (TDU)	50% Lethal Dose (TDU)
Eisenberg et al. [17]	-14.9	2.56	960	2380
Tsao & Perry [13]	-12.8	2.56	420	1046
TNO [18]	-15.3	3.02	389	841
Lees [19]	-10.7	1.99	828	2670

NFPA 1971 proximity ensembles (aluminized, TPP ≥ 50) delay burn onset by reducing effective flux.

SFPE Handbook and IOGP adjust the probit function:

$$Y = -14.9 + 2.56 \times \ln (TDU / 10^4)$$

Table 09: TDU (Geared)

TDU (Geared)	Pr	% Fatality	Gear Failure Mode	Max Time at 10 kW/m ²
1,000	2.67	1%	Minor insulation loss	~100 s
2,000	5.00	50%	TPP failure, 2nd-degree burn	~200 s
3,000	6.10	84% the gear	Catastrophic burn-through	~300 s
4,000	7.00	98%	Lethal	~400 s

Table 10: Probit Decision Matrix for Commanders

TDU (Geared)	Pr	Command Action	Rotation Frequency
<1,000	<2.67	Continue	Every 5 min
1,000–2,000	2.67–5	Rotate	Every 2 min
>2,000	>5	RETREAT	IMMEDIATE

Lethality Thresholds

Lethality is defined as 100% probability of death from thermal radiation — the ultimate failure mode in oil and gas fire suppression. NFPA 1971 (2018) sets the minimum performance baseline for protective ensembles, but lethality thresholds are derived from human physiology, probit models, and field data (CCPS, HSE UK, IOGP 434-14, API RP 521, SFPE Handbook).

Table 11: Absolute Lethality Thresholds (Unprotected)

Flux (kW/m ²)	Time to 100% Lethality	Distance (20 m Pool Fire)	Mechanism
35	20 seconds	~10 m	Instant 3rd-degree burn + shock
25	1 minute	~15 m	Clothing ignition + full-body burn
12.5	>10 minutes	~20 m	Not lethal alone

Table 11: NFPA angulation 1971 Lethality Thresholds (Geared Firefighters)

Flux (kW/m ²)	Max Time Before Gear Failure	Distance (20 m Pool Fire)	Lethal Outcome
35	<30 seconds	<10 m	Gear burn-through → 100% fatality
25	~45 seconds	~12 m	TPP collapse → lethal exposure
12.5	~2–3 minutes	~20 m	TDU > 4,000 → 98% fatality
10.0	~4 minutes	~25 m	TDU = 4,000 → lethal if not rotated

Protective ensemble for firefighting agents

“Structural firefighting protective ensembles shall NOT be used for proximity firefighting operations.” — NFPA 1971, 1.3.2

Why NFPA 1971:

There exists no international standard dedicated exclusively to protective ensembles for oil and gas firefighting. NFPA 1971 (2018 Edition) is therefore applied as the most rigorous and comprehensive standard for all types of firefighting, including structural, proximity, and hydrocarbon incidents.

Oil & gas fires (jet fires, pool fires, BLEVEs) generate sustained radiant heat fluxes > 30 kW/m² — 10× higher than building fires.

NFPA 1971 (2018) is the only consensus standard that defines two distinct protective systems:

1. Structural Fire Fighting Ensemble – for flame contact in buildings
 2. Proximity Fire Fighting Ensemble – mandatory for radiant-dominant hydrocarbon fires
- Guidelines such as IOGP 434-14, API RP 2001, HSE COMAH, and OSHA 1910.156 recommend that firefighting PPE comply with recognized standards (e.g., NFPA 1971 or EN 469) when

personnel are expected to be exposed to radiant fluxes above safe comfort limits (typically ≥ 2 kW/m²).

What NFPA 1971 Provides:

NFPA 1971 defines the **minimum performance, design, testing, and certification requirements** for protective ensembles worn during:

- Structural firefighting ensemble for interior firefighting and direct flame contact.
- Proximity firefighting ensemble for operations with **high radiant heat** but **no direct flame contact**

NFPA 1971 Requirements

NFPA 1971 (2018) establishes three core categories of requirements for protective ensembles and garments:

- Design Requirements
- Performance Requirements
- Test Methods

These apply to both structural and proximity garments.

1. Design Requirements:

It defines how every garment (structural or proximity) must be physically constructed to survive fireground stresses — before any performance testing occurs.

- Scope: Applies to all garments in the protective ensemble
- Focus: Design features that ensure fit, function, durability, and compatibility
- Proximity Garments: Enhanced design rules for radiant heat > 30 kW/m²

According to NFPA 1971 (2018), all firefighting protective ensemble elements shall be **designed, inspected, tested, certified, and labeled** by a recognized third-party organization such as **SEI or UL** to ensure compliance with the standard's performance requirements.

Each ensemble must provide **full-body protection** ("coverall concept") against heat, flame, and limited moisture penetration while maintaining **ergonomic design, comfort, and appropriate sizing** to fit the range of users in service.

The standard also specifies requirements for **materials, dimensions, and overall system integration** to ensure that all elements work together as one complete protective system.

The ensemble elements defined in this section include:

Protective Garments (Coat and Trousers)

Protective garments must be constructed of a **three-layer composite system**:

- **Outer shell:** flame- and abrasion-resistant (Nomex, Kevlar, PBI).
- **Moisture barrier:** prevents penetration of water, steam, and light chemicals.
- **Thermal barrier (liner):** provides insulation from radiant and conductive heat.

They must include a **Drag Rescue Device (DRD)** for emergency extraction of an injured firefighter and be fitted with **fluorescent and reflective trim** for visibility.

Garments must provide **complete overlap** between coat and trousers, resist **heat, flame, and mechanical stress**, and use **non-melting threads and closures**.

Protective Helmet

Helmets shall protect the head from **impact, penetration, and radiant heat**.

Each helmet must include:

- A **face shield or goggles** for eye and face protection,
- **Neck shrouds or ear covers** to protect the side and rear of the head,
- An **energy-absorbing system** inside the shell,
- A **retention (chin strap) system** to keep it secure during motion,
- **Fluorescent and reflective trim** for visibility.

For proximity firefighting, the helmet must resist **high radiant heat** and maintain **optical clarity** of the visor under exposure.

Protective Gloves

Gloves must provide **thermal and flame resistance** while maintaining **dexterity and grip**.

They are also made from **multi-layer construction** (outer shell, moisture barrier, thermal liner) and must resist **puncture, cuts, and conductive heat**.

The wrist area must overlap with the coat sleeve or include **protective wristlets** to prevent any skin exposure.

Protective Footwear

Footwear must protect against **heat, puncture, compression, and slipping**.

It includes a **heat-resistant outer sole** (tested up to 260 °C for 5 min), a **toe cap**, and a **moisture barrier** for waterproofing.

Boots must provide **electrical insulation** and remain flexible after heat aging.

Protective Hood

The hood covers **the head, neck, ears, and face areas** not protected by the helmet or SCBA mask.

It must be made from **two layers of flame-resistant fabric** (Nomex, Kevlar, or PBI blends) and maintain its **shape and elasticity** after exposure to heat and laundering.

It must not ignite, melt, or drip under test conditions and must ensure no exposed skin remains between the helmet, mask, and coat collar.

Protection Against Liquids and Contaminants

NFPA 1971 does not cover full chemical or vapor protection (these are under NFPA 1991–1994), but it requires that ensemble components provide **limited liquid protection** through the **moisture barrier**.

This barrier prevents the penetration of **water, steam, and light chemicals** and protects against **blood-borne or biological contaminants**.

The standard also requires compatibility with the **SCBA (Self-Contained Breathing Apparatus)** to maintain the integrity of protection when worn together.

2. Performance Requirements

This chapter establishes the functional performance requirements that every element of the firefighter's protective ensemble must meet to ensure a consistent level of protection,

durability, and usability during actual fire and rescue operations.

Unlike the *Design Requirements* chapter, which focuses on construction details, this part defines how the ensemble must perform under stress, through standardized testing and evaluation.

The Performance Requirements specify what the protective clothing and equipment must be capable of doing, rather than how they are built.

Each ensemble element — including the garment, helmet, gloves, footwear, hood, wristlet and moisture barrier — must demonstrate reliable performance against thermal, mechanical, and environmental hazards while maintaining comfort and mobility.

Key Aspects Covered

1. Thermal and Flame Resistance

- The standard requires that materials and composites resist ignition, melting, and heat transfer, ensuring the firefighter remains protected during exposure to flames or radiant heat.

2. Durability and Strength

- All materials, seams, and closures must retain their protective and mechanical integrity after repeated use, cleaning, and thermal exposure.
- This ensures that performance does not deteriorate under normal operational conditions.

3. Moisture and Liquid Protection

- The ensemble must prevent the penetration of water, steam, and biological contaminants through the moisture barrier.
- This protects the wearer from steam burns and exposure to harmful fluids.

4. Mechanical and Impact Protection

- Helmets, gloves, and footwear are tested for their ability to resist impact, puncture, and compression while maintaining functionality.

5. Integration and Interface Performance

- The chapter emphasizes the interaction between all parts of the ensemble.
- Overlaps between coat and trousers, gloves and sleeves, hood and helmet, and boots and pants must remain sealed, ensuring there are no gaps that expose skin during movement or when wearing the SCBA.

6. Comfort, Ergonomics, and Functionality

- The ensemble must allow adequate mobility, dexterity, and vision while minimizing heat stress and fatigue.
- Performance is linked not only to protection but also to how effectively the firefighter can perform under demanding conditions.

7. Visibility and Identification

- Reflective and fluorescent trims are required to maintain visibility under low-light and high-heat conditions.

8. Consistency and Certification

- All performance results must be verified by independent, accredited laboratories.

- Each certified item must retain a legible label showing the manufacturer, model, certification mark, and care instructions throughout its service life.

Note: The NFPA 1971 (2018) standard contains extensive technical details, specific test methods, and performance criteria that are too comprehensive to cover fully in this brief report.

The content presented here is intended as a **practical summary and guidance** to highlight the most relevant aspects for understanding and applying the standard in the context of firefighter safety and industrial fire response.

The Test Methods chapter of NFPA 1971 (2018) contains highly detailed laboratory procedures, apparatus descriptions, and measurement criteria.

Since this report serves as a practical reference, only the main principles and testing purposes are summarized here, without including the full experimental parameters and numeric standards.

Main Performance Parameters to Verify According to NFPA 1971 (2018)

Among the main standards and values defined by NFPA 1971 (2018), the following parameters shall be verified during inspection, testing, and certification to ensure protection in high-radiant-heat proximity firefighting:

Parameter	Definition	NFPA 1971 Requirement
TPP	Thermal Protective Performance – Time (seconds) to reach 2nd-degree burn at 8.4 kW/m ² (50% convective, 50% radiant).	≥ 50
Radiant Resistance	Ability to withstand extreme radiant flux without ignition, melting, dripping, or separation.	No failure @ 84 kW/m ² × 90 s
Radiant Reflectance	Percentage of incident radiant energy (800–2600 nm) reflected by the outer shell.	≥ 90 %
THL	Total Heat Loss – Rate at which heat escapes through the garment (W/m ²).	≥ 250 W/m ²
Flame Resistance	Resistance to sustained burning after direct flame exposure.	Afterflame ≤ 2 s Char length ≤ 100 mm
Shrinkage	Dimensional change after high-temperature exposure.	≤ 5 % @ 260°C × 5 min
Liquid Penetration	Resistance to penetration by liquids (water, chemicals).	No penetration (ASTM F1358, 1 hr)

Other Standards

Other international standards also cover protective ensembles for firefighting, addressing flame, heat, water, and mechanical hazards. These are used regionally or as alternatives to NFPA 1971, depending on jurisdiction and application.

Standard	Region / Scope	Year
EN 469	Europe – Protective clothing for firefighters	2020
EN 1486	Europe – Reflective protective clothing for specialized firefighting (proximity)	2007
ISO 11999-3	International – PPE for firefighters (proximity garments)	2015
ISO 11999-4	International – PPE for firefighters (interface components)	2018
ISO 11999-5	International – PPE for firefighters (helmets)	2018
ISO 11999-6	International – PPE for firefighters (gloves)	2018
ISO 11999-9	International – PPE for firefighters (footwear)	2018
AS/NZS 4824	Australia/New Zealand – Protective clothing for firefighters	2006
GB 8965.1	China – Protective clothing for firefighters	2014
CAN/CGSB-155.1	Canada – Firefighters' protective clothing	2019

Conclusion

This report has focused specifically on the thermal dimension of firefighter safety during industrial fire scenarios in the oil and gas sector — namely, the risks associated with flames, radiant heat, and high-temperature exposure. These are the most immediate and life-threatening hazards faced by firefighting teams during suppression and rescue operations.

Industrial fires, particularly pool fires, jet fires, and fireballs, generate extreme radiant fluxes that can exceed 35 kW/m^2 , leading to rapid incapacitation and fatal injuries within seconds if appropriate protective measures are not in place. The study emphasized how such radiation levels directly affect the survival time, mobility, and effectiveness of intervention teams, even when advanced equipment is used.

To address these challenges, we analyzed the protective performance requirements and design criteria of firefighter ensembles defined by international standards such as NFPA 1971. This framework establishes clear performance limits for parameters like Thermal Protective Performance (TPP), Radiant Heat Resistance, Total Heat Loss (THL), Flame Resistance, and Shrinkage Resistance. Each of these factors plays a critical role in balancing thermal insulation and physiological endurance under operational stress.

The report underlines that no garment can eliminate thermal risk; instead, the goal is to extend the safe exposure time and allow firefighters to perform critical actions before retreat becomes necessary. Protective clothing must therefore be combined with proper exposure management, distance control, and operational coordination to ensure survival during intense radiant conditions.

Ultimately, firefighter safety during industrial fires depends on an integrated system of protection — one that merges heat-resistant clothing, proper respiratory protection, and human performance management. Understanding the interaction between flame behavior, heat transfer, and human vulnerability is essential to design effective intervention strategies and minimize fatalities in oil and gas emergencies.

Finally, this report represents the first part of a broader research series. Upcoming work will focus on complementary hazards and protection systems, including:

- Smoke and toxic gas exposure and related respiratory protection systems,
- Mechanical and impact threats encountered during structural collapse or explosion response, and
- Fatigue risk and thermal stress management, addressing physiological endurance and recovery during long-duration interventions.

Together, these studies aim to provide a comprehensive understanding of firefighter safety across all major hazard domains in oil and gas fire scenarios.

References

#	Reference / Standard	Issued by	Focus / Use in Report
1	NFPA 1971 (2018 Edition)	National Fire Protection Association	Defines performance and design requirements for firefighter protective ensembles.
2	Fire and Explosion Guidance (2018)	HSE UK	Provides human vulnerability criteria and thermal radiation limits.
3	API RP 2001	American Petroleum Institute	Industrial fire protection practices and design considerations.
4	IOGP 436-14	International Association of Oil & Gas Producers	Probit models and lethality thresholds for thermal radiation.
5	GCPQRA second edition	American Institute of chemical engineering	Quantitative risk assessment framework for oil & gas fire scenarios.

